

Current and Emerging Technology Trends

IBM Data Analyst Capstone Project
Stack Overflow Developer Survey Analysis

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EXECUTIVE SUMMARY



- This report analyzes current and emerging technology trends using the Stack Overflow Developer Survey dataset.
- Web technologies dominate both current and desired usage, with HTML/CSS, JavaScript, and TypeScript leading adoption. Python shows increased future interest.
- PostgreSQL and Microsoft SQL Server remain leading database technologies across current and desired trends.
- Cloud platforms are led by Microsoft Azure and AWS in current usage, while AWS emerges as the top desired platform. Emerging providers such as Hetzner appear in future trends.
- Spring Boot, React, and ASP.NET Core lead framework adoption, with Blazor gaining future interest.
- The majority of respondents are 25–44 years old, primarily from the United States, India, and Canada, and hold Bachelor's or Master's degrees.



INTRODUCTION



- This project analyzes data from the Stack Overflow Developer Survey to identify current and emerging technology trends.
- The survey provides insights into programming languages, databases, platforms, frameworks, and developer demographics.
- The findings provide insight into evolving industry preferences and future technology adoption.
- The objective of this analysis is to compare current usage with desired technologies to uncover emerging shifts in the development landscape.



METHODOLOGY



- The dataset used for this analysis was derived from the Stack Overflow Developer Survey.
- Data cleaning and preparation were performed to organize relevant fields related to programming languages, databases, platforms, frameworks, and demographics.
- Key variables were selected to compare current technology usage with desired future preferences.
- Interactive visualizations and dashboards were created using Looker Studio to identify trends and patterns.
- Comparative analysis was conducted to evaluate differences between current and emerging technologies.



RESULTS

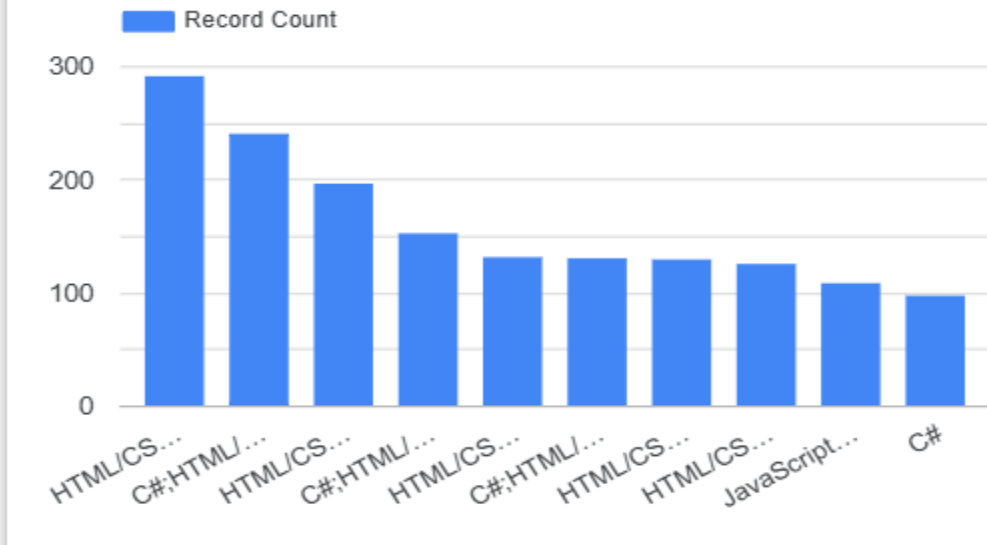
- Web technologies (HTML/CSS, JavaScript, TypeScript) dominate both current and desired usage, with Python showing increased future interest.
- PostgreSQL and Microsoft SQL Server remain leading databases across both current and desired trends.
- Microsoft Azure and AWS lead current platform usage, while AWS emerges as the most desired platform.
- Spring Boot, React, and ASP.NET Core lead framework adoption, with Blazor gaining interest in future preferences.
- Demographic data shows most respondents are ages 25–44, primarily from the United States, India, and Canada, with Bachelor's and Master's degrees most common.



PROGRAMMING LANGUAGE TRENDS

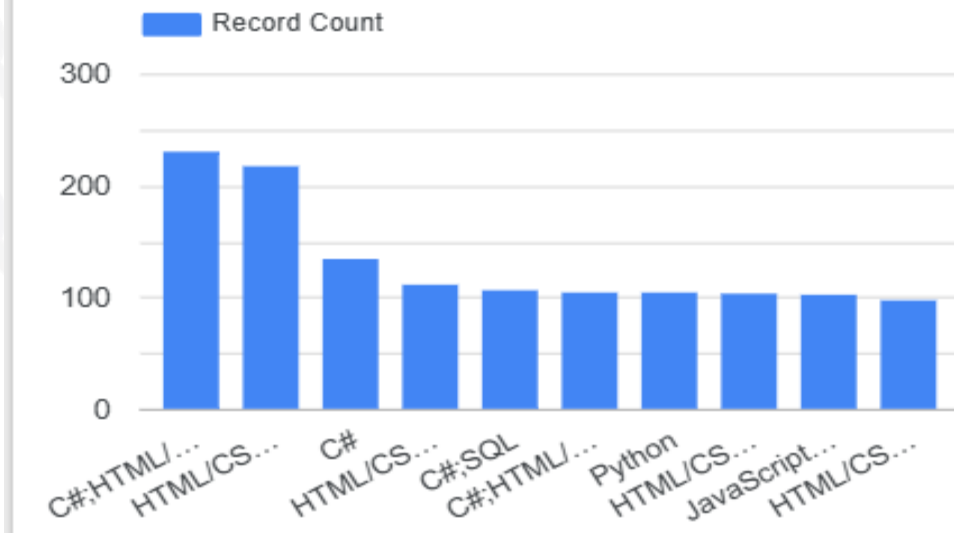
Current Year

Top 10 Programming Languages Used



Next Year

Top 10 Desired Programming La..



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Web technologies (HTML/CSS, JavaScript, and TypeScript) dominate both current and desired usage.
- Python shows increased interest in desired languages compared to current usage.
- PowerShell appears in current usage but does not show strong presence in desired technologies.

Implications

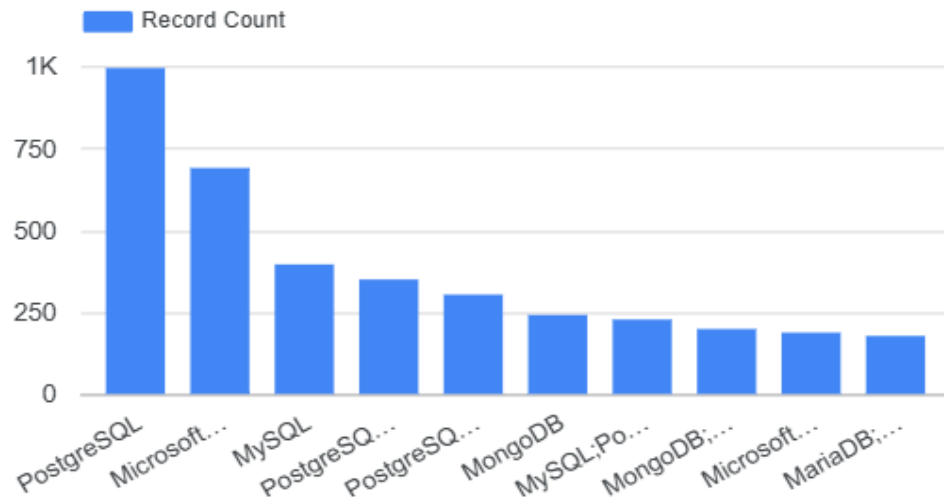
- Web development skills remain foundational and highly relevant in the industry.
- Growing interest in Python suggests continued expansion in data, automation, and backend development fields.
- Shifts away from tools like PowerShell may indicate changing priorities toward cross-platform and modern development environments.



DATABASE TRENDS

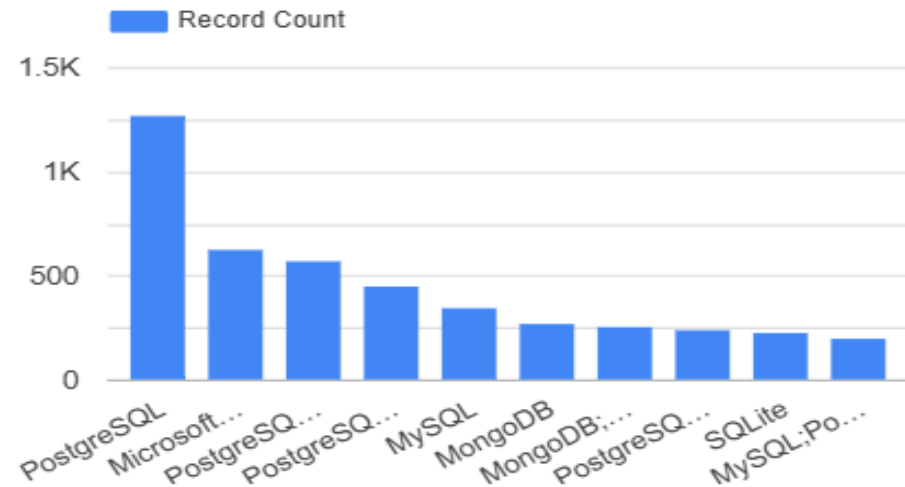
Current Year

Top 10 Databases Used



Next Year

Top 10 Databases Desired Next ...



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL and Microsoft SQL Server lead both current and desired database usage.
- MySQL, SQLite, Redis, and MongoDB maintain steady presence across current and future preferences.
- There is minimal variation between current and desired database technologies compared to programming languages.

Implications

- Organizations continue to rely on established relational database systems for stability and performance.
- Demand for SQL-based database skills is expected to remain strong in the near future.
- The stability in database preferences suggests slower technology turnover compared to other areas such as programming languages or frameworks.



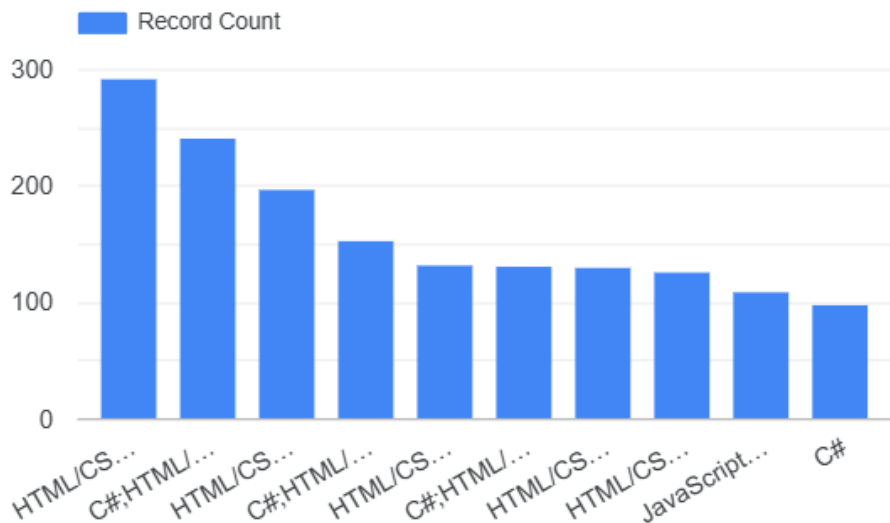
DASHBOARD



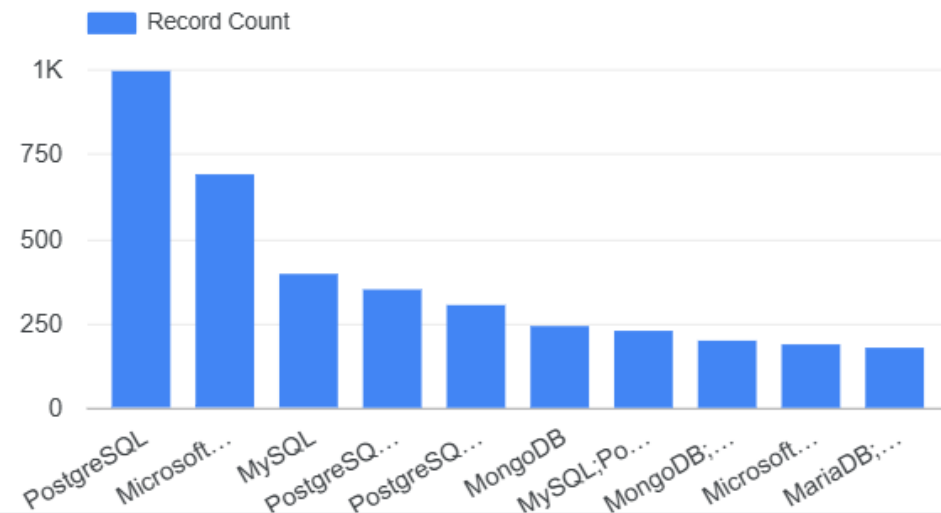
<carriegreland-wq/IBM_Capstone_Project: Most sought-after programming skills>



Top 10 Programming Languages Used



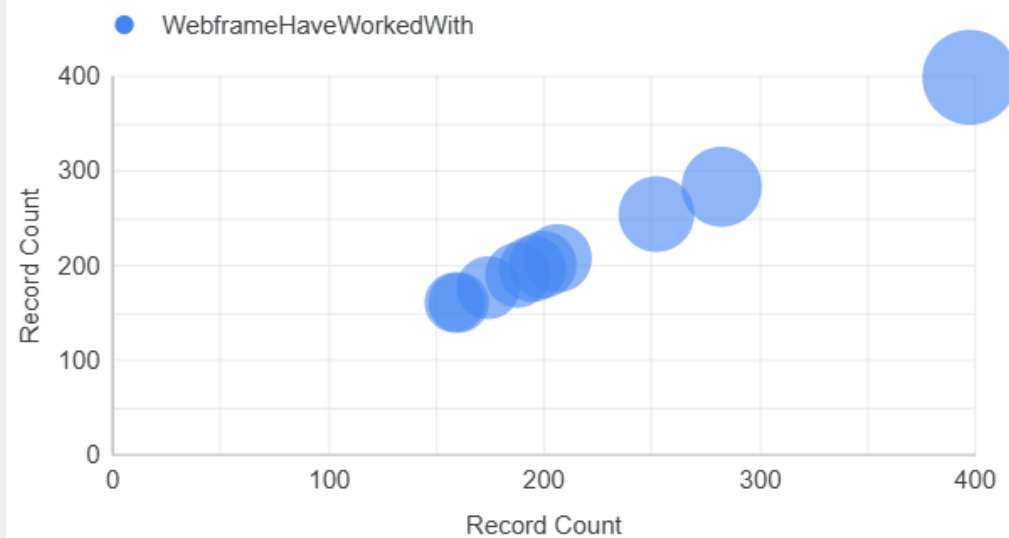
Top 10 Databases Used



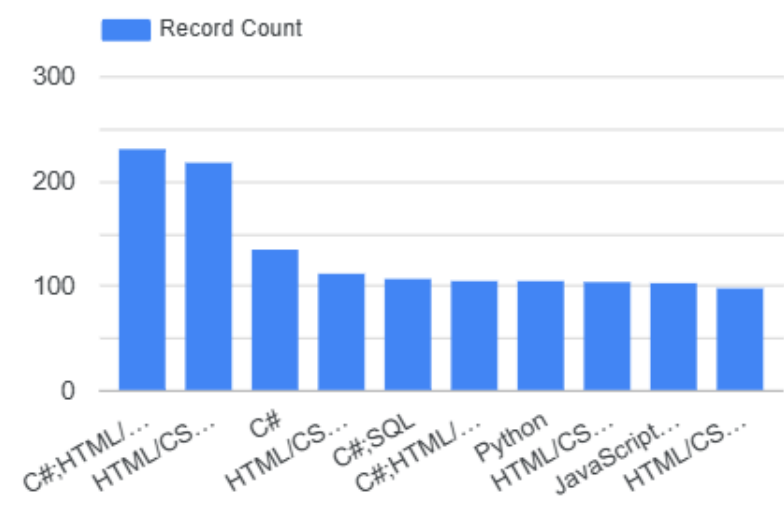
Top 10 Platforms Used



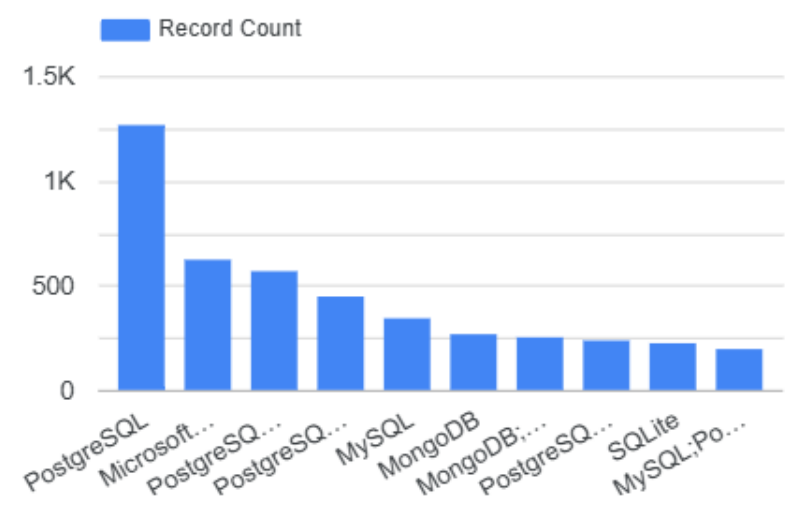
Top 10 Web Frameworks Used



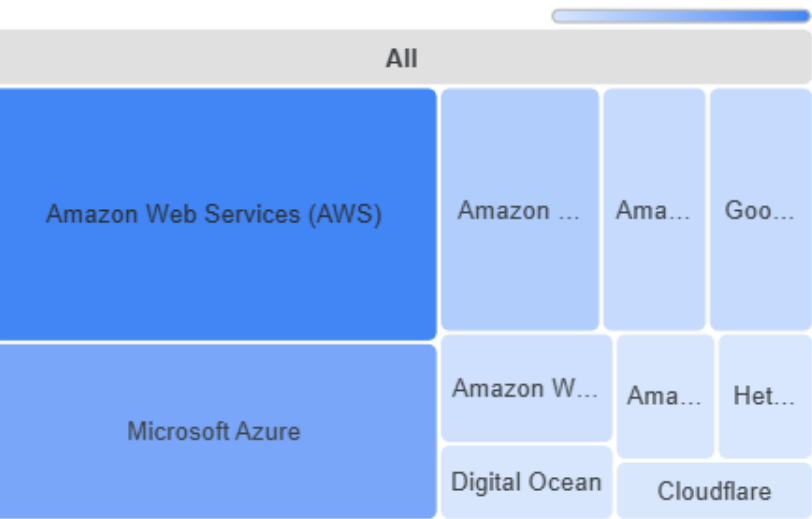
Top 10 Desired Programming La...



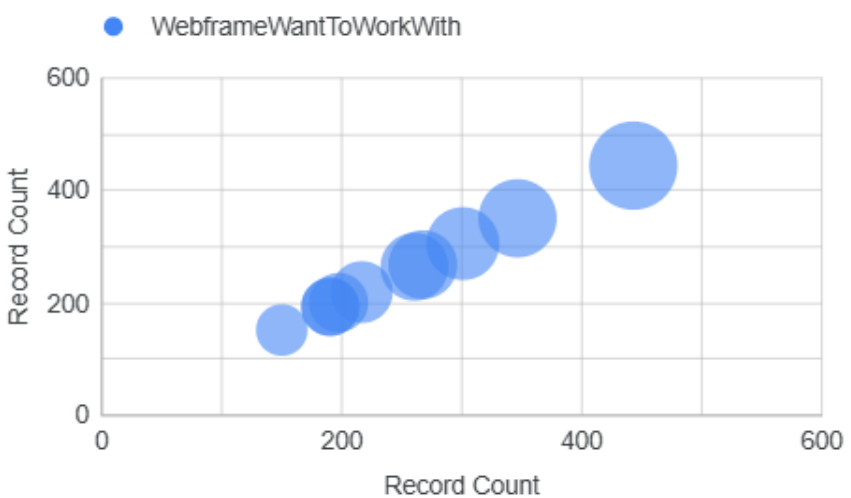
Top 10 Databases Desired Next ...



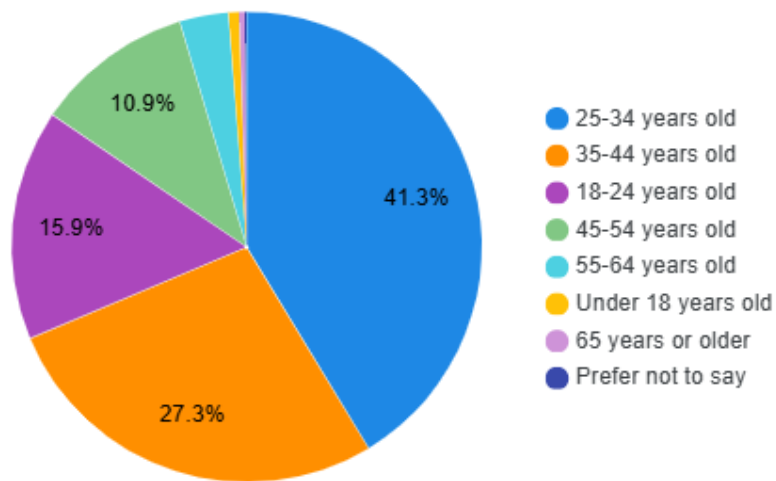
Top 10 Desired Platforms



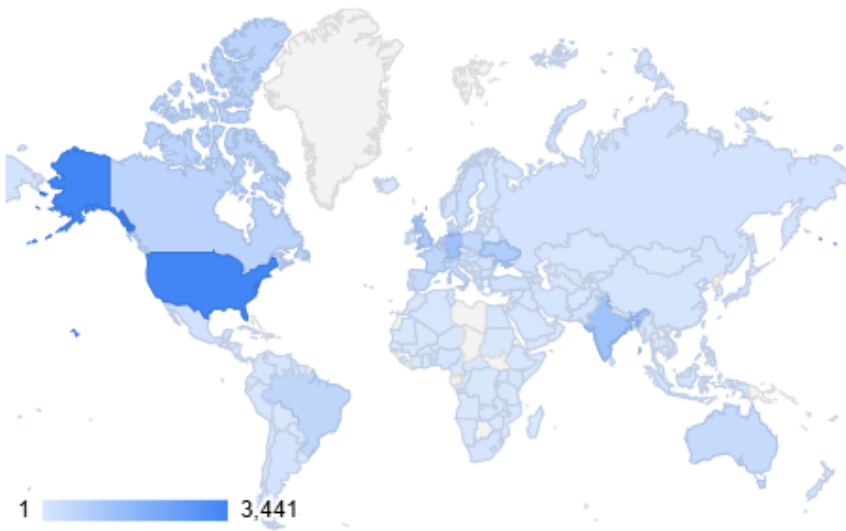
Top 10 Desired Web Frameworks



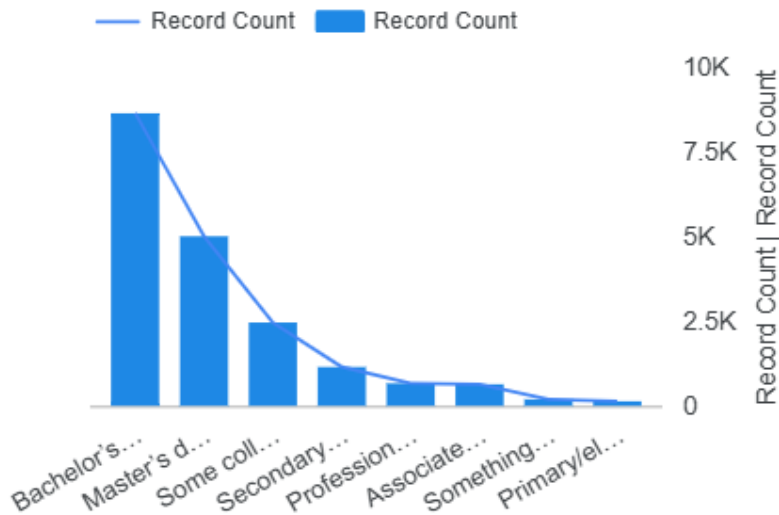
Respondents by Age



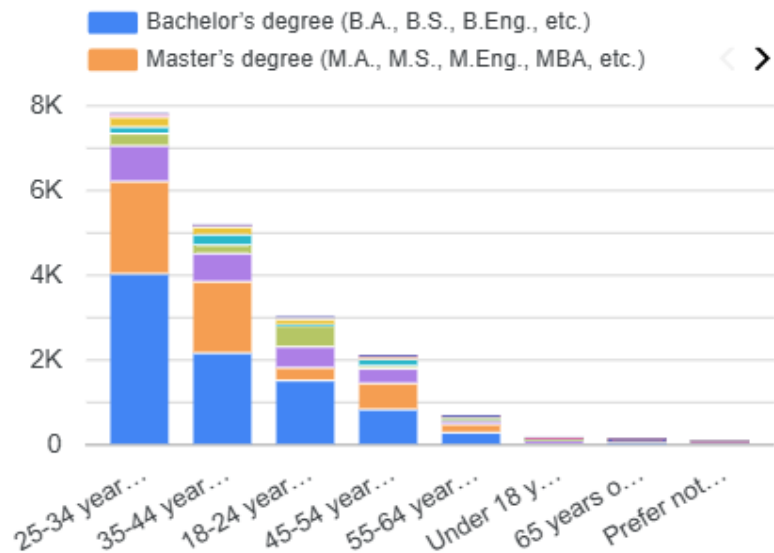
Respondent Count by Country



Respondent Distribution by Edu...



Respondent Count by Age, Clas...



DISCUSSION



- The growing interest in Python may reflect expanding demand in data science, automation, backend development, and AI-driven applications.
- The rise of AI and machine learning technologies may be influencing increased interest in languages such as Python, which are widely used in AI development and data-driven systems.
- The continued dominance of web technologies (HTML/CSS, JavaScript, TypeScript) reinforces the central role of browser-based and full-stack development in modern software ecosystems.
- Stability in database preferences suggests organizations prioritize reliability and long-term infrastructure investments over rapid change.
- The competitive positioning of AWS and Microsoft Azure indicates continued enterprise migration to cloud-first architectures.
- Emerging technologies such as Blazor and increased interest in modern frameworks may reflect demand for more efficient, scalable development solutions.
- Shifts away from tools like PowerShell could signal movement toward cross-platform development environments and containerized/cloud-native workflows.



OVERALL FINDINGS & IMPLICATIONS

Findings

- Web and cloud technologies dominate both current usage and future interest.
- Core infrastructure technologies such as relational databases show strong stability over time.
- Emerging interest in Python and modern frameworks suggests alignment with data-driven and AI-influenced development trends.

Implications

- Developers should prioritize web, cloud, and cross-platform skills to remain competitive.
- Organizations are likely to continue investing in stable infrastructure while gradually adopting emerging technologies.
- The influence of AI and modern development practices may accelerate demand for scalable, data-centric technology stacks.



CONCLUSION



- The development ecosystem continues shifting toward cloud-first, web-centered architectures.
- Increasing interest in Python signals expanding influence of AI, automation, and data-driven technologies.
- Stable database preferences highlight the importance of reliable infrastructure amid ongoing innovation.
- Emerging frameworks and platforms reflect modernization of development workflows and scalable application design.
- The convergence of cloud computing, AI, and modern frameworks is likely to shape the next phase of industry growth.



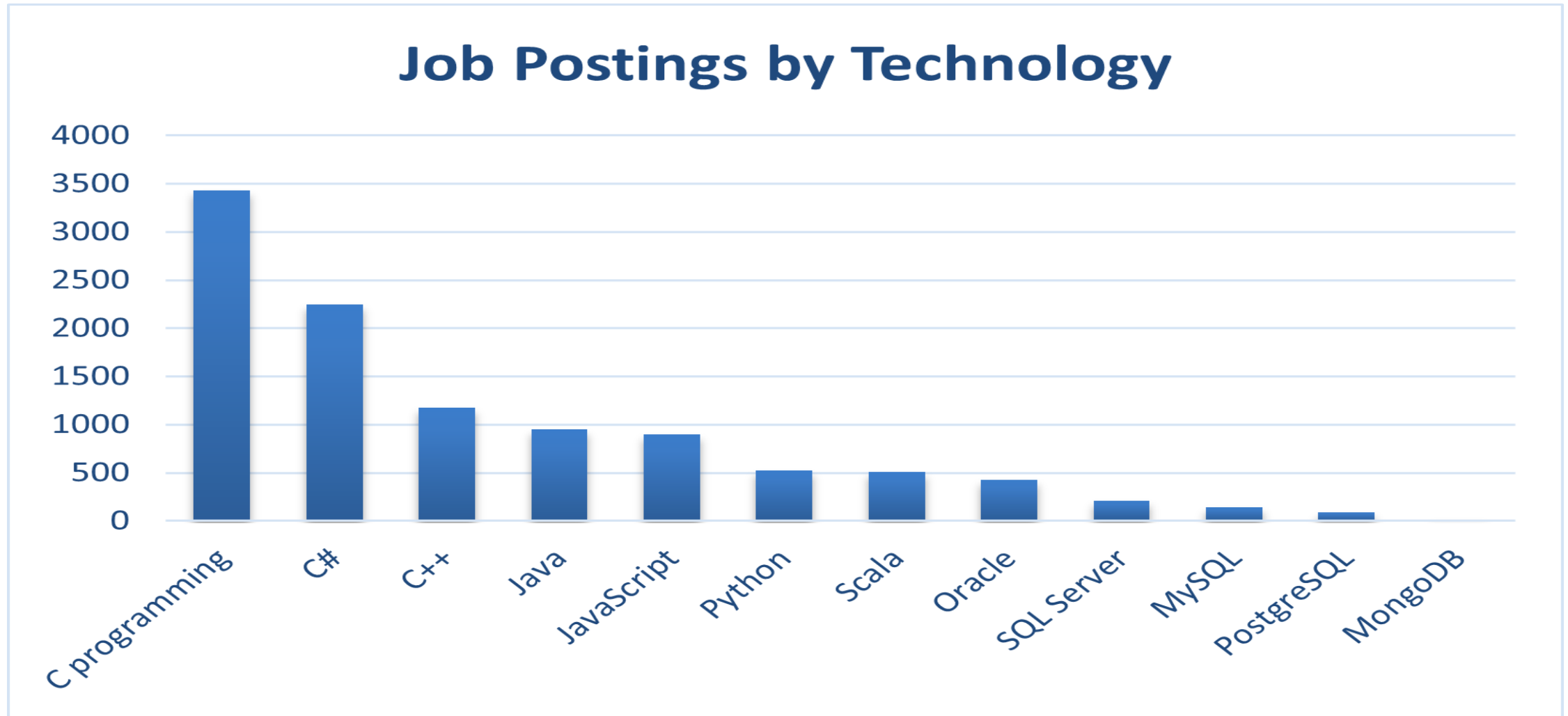
APPENDIX



- No additional supplementary materials were created beyond the dashboard visualizations.



JOB POSTINGS



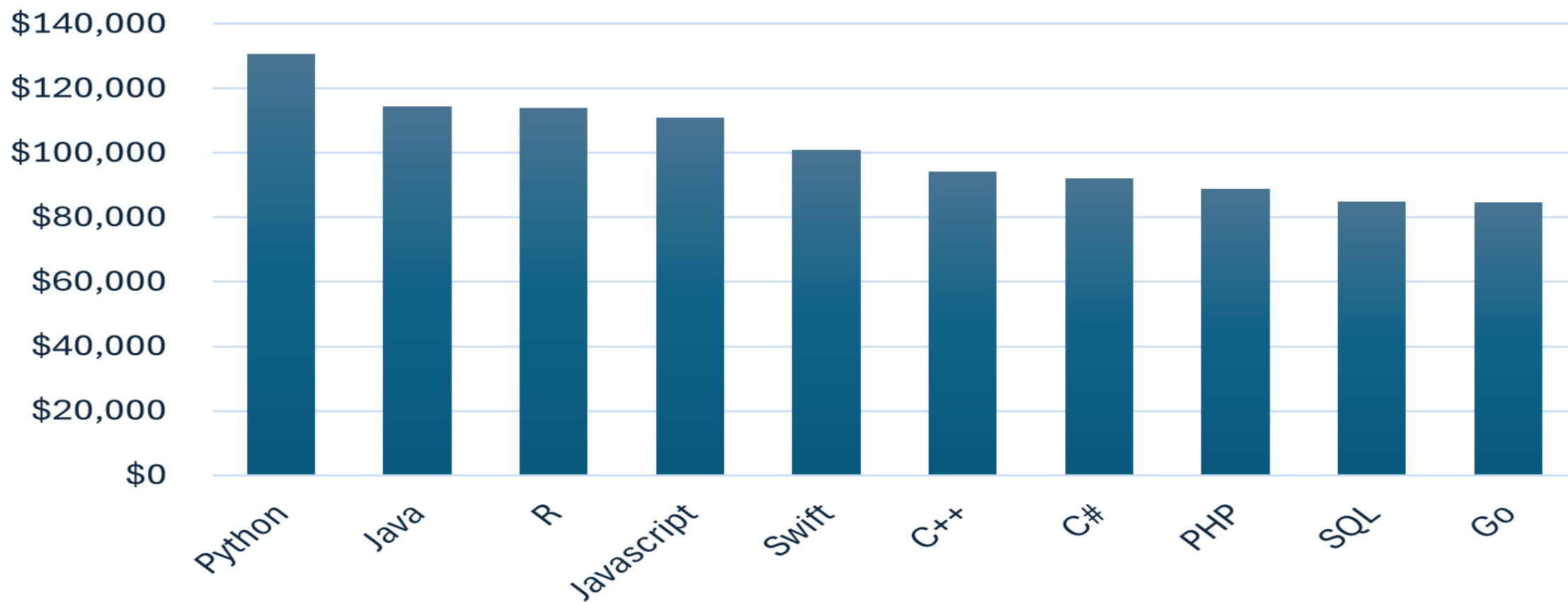
JOB MARKET INSIGHTS

- Java and JavaScript show the highest demand among analyzed technologies.
- Python maintains strong job market presence, aligning with growing interest in AI and data-driven development.
- Traditional languages such as C and PostgreSQL show lower relative demand in this dataset.



POPULAR LANGUAGES

Average Annual Salary



Salary Insights & Industry Implications

- Python commands the highest average salary among analyzed languages.
- Java and R also demonstrate strong compensation levels, reflecting enterprise and data-oriented demand.
- High salaries for Python and R further reinforce the growing influence of AI, machine learning, and analytics fields.
- Compensation trends align with survey findings showing increased interest in Python.

Final Summary

- Survey data, job demand, and salary trends indicate a converging industry direction.
- Python emerges as a leading technology across interest, job postings, and compensation — reflecting growth in AI, automation, and data-driven development.
- Web and cloud platforms continue driving scalable, cloud-first architectures.
- The data suggests strategic evolution — where stable infrastructure remains foundational while innovation accelerates in AI and modern frameworks.

